

Code Spaces Ransomware Attack: A Cybersecurity and IT Ethics Case Study



A Project Documentation
Presented to
The University of Mindanao
College of Computing
Education

In Partial Fulfillment
of the Requirements in IT17
1st Semester, 1st Term S.Y. 2025-2026

Christian Kerby
Navallo
Niel John Adlawan
Vincent Ray Toylo
Jasper Ondangan
Benz Carl Vale

November 2025

Table of Contents

ABSTRACT	3
INTRODUCTION	3
CASE DESCRIPTION.....	4
ANALYSIS.....	5
A. Compliance / Law / Policy Involved.....	5
B. Risks and Liabilities	5
C. Accountability and Responsibilities	6
D. Ethical Issues	7
LESSONS LEARNED	8
REFERENCES	9
APPENDICES	9

ABSTRACT

Cybersecurity threats have grown rapidly as organizations increasingly rely on cloud computing and online infrastructure. One notable incident that highlights the risks of poor security management is the Code Spaces ransomware attack in 2014. Code Spaces was a cloud-based source code hosting and project management service used by software developers worldwide. The company relied heavily on Amazon Web Services (AWS) to store its data and infrastructure.

In June 2014, an attacker gained access to the company's cloud control panel and attempted to extort money by threatening to delete customer data and company resources. When Code Spaces refused to pay the ransom, the attacker carried out the threat by deleting large portions of the company's data, backups, and server configurations. The attack ultimately forced the company to shut down its operations permanently.

This case study analyzes the ethical, technical, and legal aspects of the Code Spaces incident. It explores how inadequate security controls, poor access management, and insufficient backup strategies contributed to the disaster. The study also evaluates the responsibilities of organizations in protecting customer data, the ethical implications of cyber extortion, and the lessons that modern IT professionals can learn from this event. Understanding this case is important for strengthening cybersecurity practices and reinforcing ethical responsibility in the field of information technology.

INTRODUCTION

The rapid growth of cloud computing has allowed companies to store data, host applications, and manage services more efficiently. However, this convenience also introduces new security risks. When organizations rely heavily on cloud infrastructure without implementing strong security practices, they may expose themselves to severe cyberattacks.

One example of such a risk is ransomware. Ransomware is a form of malicious activity in which attackers gain unauthorized access to systems and demand payment in exchange for restoring data or preventing damage. These attacks can lead to financial loss, operational disruption, and loss of trust from customers.

In 2014, a small technology company called Code Spaces became one of the most dramatic examples of how ransomware can destroy a business. Code Spaces provided cloud-based development tools and source code hosting services for clients worldwide. The company stored its entire infrastructure on Amazon Web Services, which included application servers, databases, backups, and customer files.

The attack began when a cybercriminal obtained access to Code Spaces' AWS control panel. The attacker attempted to extort the company by threatening to delete its data unless a ransom was paid. When the company refused to comply, the attacker proceeded to delete servers, backups, and storage systems.

Unlike many cyberattacks where companies recover after restoring backups, Code Spaces was unable to fully recover its infrastructure. The damage was so extensive that the company announced it could no longer continue operations. Within days, Code Spaces shut down permanently.

This incident highlights the importance of cybersecurity preparedness, responsible data management, and ethical practices in the technology industry. It also raises questions about accountability, risk management, and the ethical responsibilities of both organizations and cybersecurity professionals.

CASE DESCRIPTION

Code Spaces was founded as a cloud-based service designed to help software developers collaborate and manage code projects online. The company relied almost entirely on Amazon Web Services to operate its platform. This included hosting servers, storing data, managing backups, and running development tools.

On June 17, 2014, an attacker gained access to Code Spaces' AWS control panel. This control panel allowed administrators to manage all of the company's servers, storage systems, and backups. Because of the level of access obtained, the attacker had control over nearly every part of the company's infrastructure.

After gaining access, the attacker contacted Code Spaces and demanded a ransom payment. The attacker threatened to delete the company's data and servers if the ransom was not paid. Instead of complying, Code Spaces attempted to regain control of its systems and remove the attacker's access.

However, during this process the attacker began deleting critical resources from the AWS environment. This included:

- Virtual machines running the company's services
- Databases containing customer information
- Backups stored for recovery purposes
- Storage buckets containing application files and user data

The deletion of these resources caused major disruptions. Code Spaces' website and services stopped functioning, and customers could no longer access their stored projects or development tools.

Timeline of events:

June 17, 2014 – Unauthorized access to the AWS control panel is detected
June 18, 2014 – The attacker demands ransom payment
June 18–19, 2014 – Code Spaces attempts to regain control of the account
June 19, 2014 – The attacker deletes servers, backups, and storage data
June 20, 2014 – Company announces severe data loss
June 23, 2014 – Code Spaces officially shuts down operations

The consequences were severe. Large portions of customer data were permanently lost, and the company's infrastructure could not be restored. Because backups were also deleted, recovery became nearly impossible.

Immediate consequences included:

- Service outages for all customers
- Loss of project files and development data
- Disruption of business operations

Long-term consequences included:

- Permanent closure of the company
- Loss of jobs for employees
- Loss of trust in cloud-based services for some users
- Increased awareness of cloud security risks

The Code Spaces attack became an important case study in cybersecurity because it showed how a single compromised administrative account could lead to the complete destruction of a company.

ANALYSIS

A. Compliance / Law / Policy Involved

Several legal and policy issues were relevant to the Code Spaces incident. Organizations that manage user data are expected to follow security standards and industry best practices to protect that information.

Relevant policies and regulations include:

- Data protection regulations requiring companies to secure customer information
- Information security management standards such as ISO 27001
- Corporate cybersecurity policies governing access control and authentication
- Cloud security best practices recommended by cloud service providers

In this case, the attack exposed weaknesses in security controls related to account access and cloud management. One critical issue was the apparent lack of strong authentication mechanisms for administrative accounts.

Security policies often require multi-factor authentication (MFA) to protect sensitive systems. MFA adds an additional layer of security by requiring more than just a password to access accounts. If MFA had been properly implemented, the attacker might not have been able to gain access so easily.

Another compliance concern involved backup management. Companies are expected to maintain reliable backups that are protected from unauthorized deletion. In the Code Spaces case, backups were stored in the same environment as the primary systems, allowing the attacker to delete them as well.

Because Code Spaces handled customer data, there was also an ethical and legal expectation to ensure that proper safeguards were in place.

B. Risks and

Liabilities

Technical Risks

- Unauthorized access to cloud infrastructure
- Deletion of servers and databases
- Loss of backup systems
- Permanent loss of customer data

Operational Risks

- Complete shutdown of company services
- Disruption of client projects and workflows
- Loss of company infrastructure

Financial and Reputational Impact

- Loss of revenue from service outages
- Cost of attempting recovery
- Permanent closure of the business

Reputational Risks

- Loss of customer trust
- Damage to the company's brand image
- Reduced confidence in similar cloud services

Potential liabilities also arise when customer data is lost or compromised. Clients who rely on hosted platforms may experience financial damage if their data disappears. Although Code Spaces was also a victim of the attack, organizations still have a responsibility to protect their systems from foreseeable risks.

C. Accountability and

Responsibilities

Several parties played roles in this incident, either directly or indirectly.

The primary responsibility lies with the attacker who intentionally accessed the company's systems without authorization and attempted to extort money. This behavior is clearly illegal and unethical.

However, organizations also have responsibilities when managing digital systems and customer data. Code Spaces had the duty to implement strong security controls to prevent unauthorized access.

Areas where responsibility may have been lacking include:

- Insufficient access control mechanisms
- Lack of multi-factor authentication for critical accounts
- Backup systems not protected from deletion
- Over-reliance on a single cloud environment

Users also share some responsibility for understanding the risks associated with online services. However, most customers depend on service providers to implement strong security measures.

Although law enforcement agencies investigate cybercrime, identifying and prosecuting attackers can be difficult, especially when attacks cross international boundaries.

D. Ethical Issues

The Code Spaces attack raises several important ethical concerns in information technology.

Cyber Extortion

The attacker attempted to force the company to pay money by threatening to destroy data. This is a clear example of cyber extortion and violates ethical principles related to fairness, responsibility, and respect for others.

Intentional Harm

Deleting company infrastructure and customer data caused serious damage to businesses and individuals. Ethical computing principles emphasize that technology should be used to benefit society rather than cause harm.

Negligence in Security

Organizations that store user data have an ethical obligation to protect that information. Failure to implement proper security practices can be considered negligence.

Data Responsibility

Companies handling digital assets must ensure the availability and integrity of data. Ethical IT professionals are expected to design systems that minimize risks and protect users.

Professional Integrity

IT professionals are expected to follow codes of ethics such as those promoted by professional organizations in computing and cybersecurity. These codes emphasize honesty, responsibility, and protection of public interests.

LESSONS LEARNED

The Code Spaces ransomware attack demonstrates how critical cybersecurity practices are for modern organizations. A single security breach can lead to catastrophic consequences if systems are not designed with strong protections and recovery plans.

One important lesson is the need for strong authentication and access control. Administrative accounts should always be protected using multi-factor authentication to reduce the risk of unauthorized access.

Another key lesson is the importance of backup strategies. Backups should be stored in separate and secure environments so they cannot be deleted by attackers who compromise the main system.

Organizations must also implement monitoring systems that detect suspicious activities quickly. Early detection can prevent attackers from causing widespread damage.

Employee training is also important. Staff members responsible for managing infrastructure must understand cloud security practices and risk management.

From an ethical perspective, this case reminds IT professionals of their responsibility to design secure systems and protect user data. Ethical decision-making and responsible system management are essential for maintaining trust in digital technologies.

Finally, governments and organizations must continue improving cybersecurity laws, policies, and international cooperation to combat cybercrime effectively.

The Code Spaces case serves as a powerful reminder that cybersecurity failures can destroy not only data but entire organizations.

REFERENCES

- [1] Code Spaces (2014). Company Statement on Security Incident.
- [2] Krebs, B. (2014). "Hacked Code Spaces Forced to Close Its Doors." Krebs on Security.

- [3] National Institute of Standards and Technology (NIST). Cybersecurity Framework.
- [4] International Organization for Standardization. ISO/IEC 27001 Information Security Management.

APPENDICES

[FILTER ARTICLES](#)[SUBSCRIBE](#) [SEARCH](#) 

January 2020 after being [hit](#) by a cyber attack on New Year's Eve.



The perpetrators, ransomware gang REvil, [claimed](#) they had gained access to the firm's network six months previously and demanded that Travelex pay a \$6m (£4.6m) ransom. The attack was devastating, causing a month of disruption with staff unable to use computers to keep track of currency trading.

It took until 17 January for the company to finally announce that services were back up and running. The attack impacted high-profile clients too, including Barclays, Asda and Royal Bank of Scotland.

From there, things continued to tumble downhill. First, Travelex put itself up for sale and then in August, it went into administration with 1,300 of its employees losing their jobs. Its creditors then bought it in a lucky escape for the beleaguered firm.

"While this breach did not solely contribute to the fact that the company entered administration, it



[supporting links](#)

[supporting link](#)